

Gold Future/ETF Analysis

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Table 1: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold from 12/22/2008 to 7/14/2014.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.94314	2.419×10^{-5}	0.80947	0.00530
2-Month	0.94301	$\approx 10^{-5}$	0.80911	0.00530
6-Month	0.94348	3.180×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.233×10^{-5}	0.80984	0.00529

Table 2: A summary of the slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

Table 3: A summary of the weights and in/out of sample RMSEs for portfolios of 1, 2, and 3 futures contracts.

Portfolio		w_0	w_1	w_2	w_3	RMSE (in)	RMSE (out)
1 Futures	1-m	-0.16053	1.16053			0.00285	0.00427
	2-m	-0.17993	1.17993			0.00299	0.00450
	6-m	-0.18732	1.18732			0.00295	0.00541
	12-m	-0.06259	1.06259			0.00483	0.00684
2 Futures	1-m, 2-m	-0.16936	0.85695	0.31241		0.00283	0.00434
	1-m, 6-m	-0.17736	0.71410	0.46326		0.00279	0.00450
	1-m, 12-m	-0.19057	1.01853	0.17204		0.00279	0.00445
	2-m, 6-m	-0.19181	0.53762	0.65419		0.00285	0.00478
	2-m, 12-m	-0.20928	1.01944	0.18983		0.00291	0.00473
	6-m, 12-m	-0.21749	1.04815	0.16933		0.00287	0.00504
3 Futures	1-m, 2-m, 6-m	-0.18008	0.60862	0.14966	0.42180	0.00278	0.00450
	1-m, 2-m, 12-m	-0.19688	0.79086	0.24270	0.16333	0.00278	0.00450
	1-m, 6-m, 12-m	-0.20496	0.64458	0.41532	0.14506	0.00274	0.00464
	2-m, 6-m, 12-m	-0.21907	0.47364	0.59770	0.14773	0.00280	0.00488